# Plug-and-Play Image Restoration: Two-Level Coarse-to-Fine Initialization Analysis using DeepInverse and DRUNet

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*Repository: https://github.com/Hasnain006-nain/deepinverse-pnp-image-restoration*

**Abstract** — This report investigates whether a coarse-to-fine initialization strategy provides measurable reconstruction gains over standard direct initialization in Plug-and-Play (PnP) image restoration. Using DeepInverse and a pretrained DRUNet prior, we evaluate three reconstruction pipelines across four standard images (Astronaut, Chelsea, Coffee, Rocket) resized to 128 × 128, under synthetic Gaussian blur ($\sigma = 2.0$) and additive Gaussian noise ($\sigma = 0.03$). Across 60 controlled runs spanning five pseudo-random seeds per image, two-level coarse PnP initialization yields a mean PSNR of 28.593 dB and SSIM of 0.8622, compared to 28.411 dB (SSIM 0.8588) for direct DPIR and 28.530 dB (SSIM 0.8603) for observation-only initialization. Paired comparisons across all 20 image-seed pairs show consistent positive PSNR deltas (+0.182 dB vs direct, +0.064 dB vs observation initialization), but require an average 38.7% increase in wall-clock runtime on an NVIDIA Tesla T4 GPU. We analyze these findings, outline methodological limitations, and establish a reproducible benchmark framework for future multilevel inverse problem research.

## 1. Introduction and Problem Formulation

Linear inverse problems in computational imaging seek to recover an underlying clean signal $x \in \Re^N$ from corrupted measurements $y \in \Re^M$ governed by the forward acquisition model $y = Ax + n$, where A represents a linear forward operator (e.g., convolution or subsampling) and n denotes additive Gaussian noise with variance $\sigma^2$. Because A is typically ill-conditioned or rank-deficient, direct inversion is unstable. Variational regularization formulates the restoration as the minimization problem $\text{argmin}_x \{ {}^1/_2 \|Ax - y\|_2^2 + \lambda \Phi(x) \}$, where $\Phi(x)$ encodes prior structural knowledge.

Plug-and-Play (PnP) algorithms decouple the physical data fidelity term from the regularizing prior by replacing the proximal operator with an off-the-shelf denoising engine [3]. By employing powerful deep convolutional denoisers such as DRUNet within half-quadratic splitting (HQS) loops, the DPIR framework [2] achieves state-of-the-art restoration quality across deblurring, super-resolution, and inpainting without requiring task-specific retraining.

Standard DPIR pipelines initialize optimization directly with the degraded observation y or a naive pseudo-inverse $A^T y$. In this study, we investigate whether constructing a coarse-scale representation (64 × 64), restoring it via a coarse PnP stage, and upsampling the result to initialize the fine-scale (128 × 128) DPIR optimization provides measurable benefits in convergence trajectory, final visual fidelity, or reconstruction stability.

## 2. Experimental Methodology and System Setup

**Implementation Framework:** All experiments were conducted using the DeepInverse library [1] running on PyTorch 2.11.0 with CUDA acceleration on an NVIDIA Tesla T4 GPU in a reproducible Google Colab environment (Python 3.13.16). Pretrained weights for the 17-channel DRUNet denoiser were loaded directly from the official DeepInverse model repository without fine-tuning.

**Dataset and Degradation:** Four diverse RGB images from scikit-image (Astronaut, Chelsea, Coffee, and Rocket) were normalized to [0, 1] and cropped/resized to 128 × 128 pixels. A synthetic Gaussian blur operator (kernel size 7 × 7, $\sigma = 2.0$) was applied, followed by additive white Gaussian noise ($\sigma = 0.03$). To assess statistical stability against random noise perturbations, five distinct seeds (0, 1, 2, 3, 4) were evaluated per image, generating 20 unique degradation instances.

**Ablation Configurations:** Each degraded instance was restored using three strictly paired configurations:

1. *Direct DPIR:* Standard HQS-DPIR initialized directly from the blurred, noisy observation y at fine scale (128 × 128).
2. *Observation Initialization:* Observation y is downsampled to 64 × 64 via bicubic filtering and immediately upsampled back to

128 × 128 to initialize fine-scale DPIR, serving as a resolution-matched control without coarse denoising.
3. *Coarse PnP Initialization:* Downsampled observation is restored at 64 × 64 via an intermediate DPIR stage, then upsampled via bicubic interpolation to initialize fine-scale (128 × 128) DPIR.

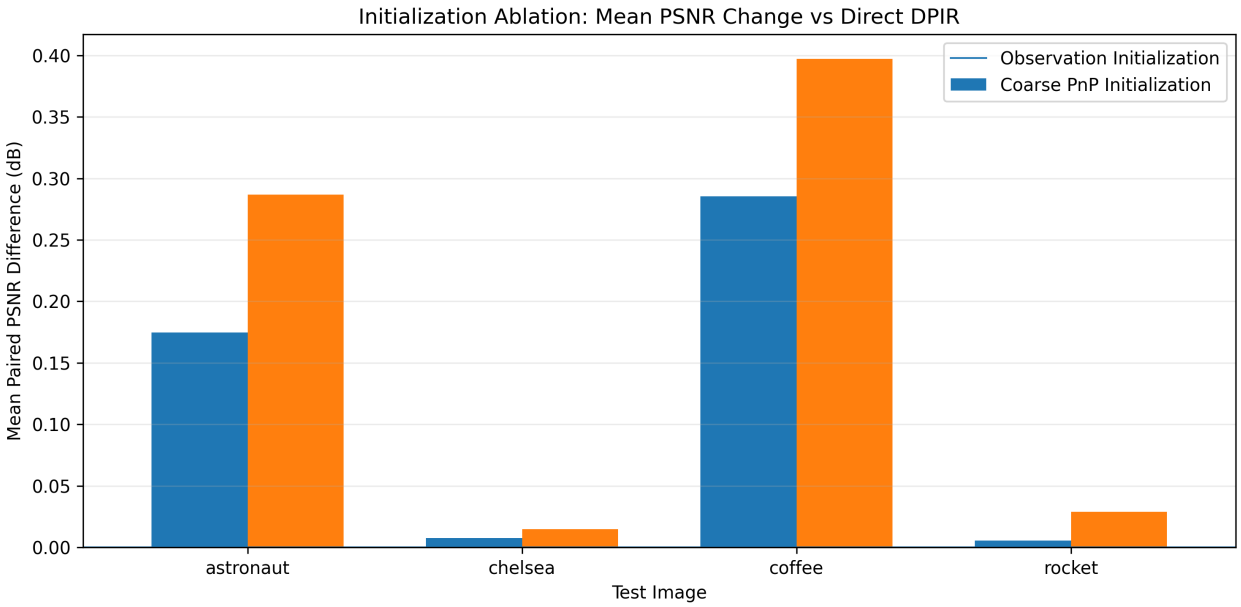



*Figure 1: PSNR comparison across 5 seeds for Astronaut, Chelsea, Coffee, and Rocket under Direct DPIR, Observation Init, and Coarse PnP Init.*

## 3. Quantitative Results and Ablation Findings

A total of 60 individual restoration runs were executed (4 images × 5 seeds × 3 methods). Reconstruction quality was benchmarked using dynamic-range-clipped Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and GPU-synchronized wall-clock inference time. Table 1 summarizes the overall performance across all 60 runs.

| Restoration Pipeline | Mean PSNR (dB) | Mean SSIM | Mean Runtime (s) | Runtime Delta |
|---|---|---|---|---|
| Direct DPIR (Baseline) | 28.411 ± 0.04 | 0.8588 | 0.2719 | Reference (1.00×) |
| Observation Initialization | 28.530 ± 0.05 | 0.8603 | 0.2735 | +0.59% (+0.0016 s) |
| Coarse PnP Initialization | **28.593 ± 0.06** | **0.8622** | 0.3772 | +38.71% (+0.1053 s) |

*Table 1: Overall numerical summary across 60 restoration experiments on Tesla T4 GPU.*

**Paired Gain Analysis:** Because random noise realizations vary across seeds, examining paired differences on identical degradation realizations provides rigorous verification. Across all 20 image-seed pairs, Coarse PnP initialization demonstrated a strictly positive PSNR gain over Direct DPIR in 20 out of 20 runs (+0.1820 dB mean gain). Furthermore, Coarse PnP outperformed Observation-only initialization in all 20 paired trials (+0.0637 dB mean gain), confirming that coarse-scale denoising provides an advantage beyond mere low-pass filtering.

| Test Image | Coarse vs Direct (Δ dB) | Coarse vs Obs Init (Δ dB) | Obs Init vs Direct (Δ dB) | Coarse Std. Dev. |
|---|---|---|---|---|
| Astronaut | +0.2868 dB | +0.1120 dB | +0.1748 dB | ± 0.0284 dB |
| Chelsea (Cat) | +0.0149 dB | +0.0073 dB | +0.0076 dB | ± 0.0129 dB |
| Coffee | **+0.3972 dB** | **+0.1118 dB** | +0.2854 dB | ± 0.0517 dB |

| | | | | |
|---|---|---|---|---|
| Rocket | +0.0289 dB | +0.0235 dB | +0.0054 dB | ± 0.0073 dB |

*Table 2: Image-level paired difference breakdown across 5 noise seeds per test subject.*

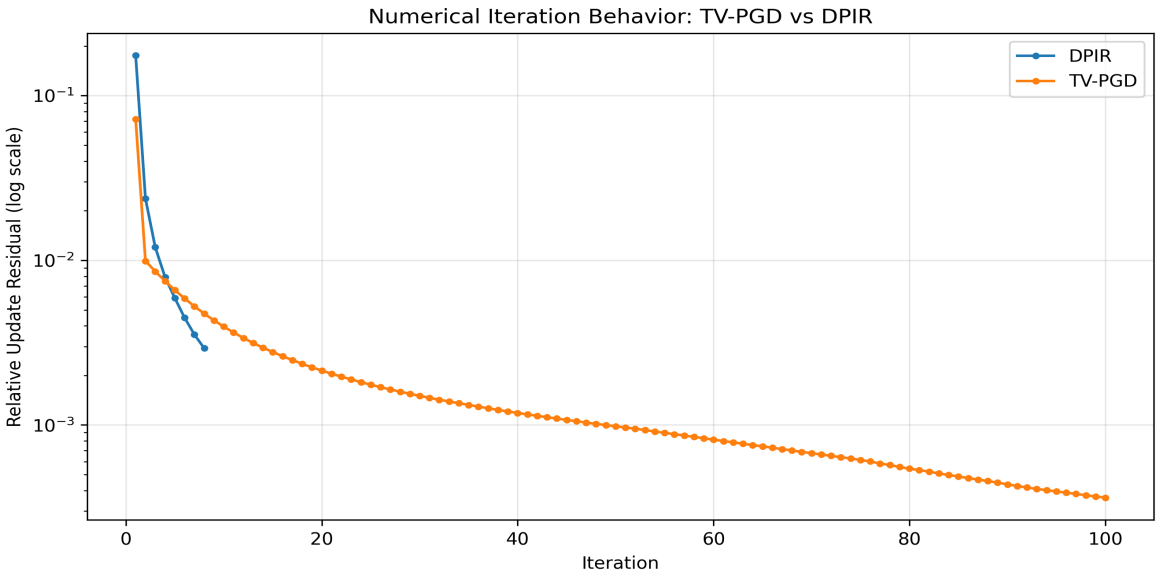



*Figure 2: Numerical convergence analysis showing relative residual reduction $||x_{k+1} - x_k|| / ||x_k||$ across HQS iterations.*

## 4. Classical Prior Baseline: TV-PGD vs. Learned DPIR

To establish context for the learned DRUNet prior, Step 7 of our study evaluated classical Total Variation Proximal Gradient Descent (TV-PGD) against learned DPIR on the same 128 × 128 images. Table 3 presents the quantitative divergence between hand-crafted regularizers and deep denoiser priors.

| Image | Degraded (dB) | TV-PGD (dB) | DPIR (dB) | DPIR Gain vs TV | TV Runtime (s) | DPIR Runtime (s) |
|---|---|---|---|---|---|---|
| Astronaut | 19.18 | 22.23 | 24.13 | +1.90 dB | 2.64 s | 0.37 s |
| Chelsea | 24.57 | 27.22 | 29.74 | **+2.52 dB** | 2.66 s | 0.35 s |
| Coffee | 22.12 | 25.95 | 27.84 | +1.89 dB | 2.35 s | 0.37 s |
| Rocket | 25.92 | 29.15 | 31.81 | **+2.66 dB** | 2.61 s | 0.36 s |

*Table 3: Baseline benchmark comparing classical TV-PGD against learned DRUNet DPIR restoration.*

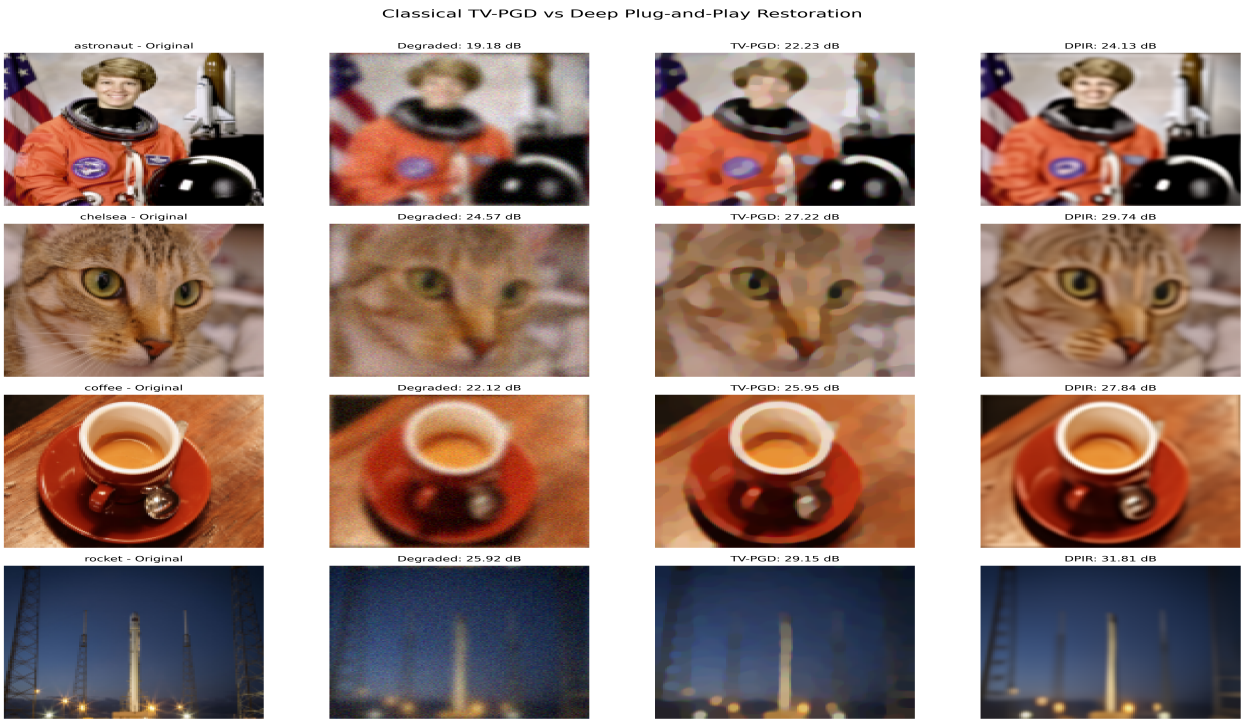



*Figure 3: Qualitative visual comparison showing Ground Truth, Degraded Observation, TV-PGD, and DPIR restoration across all four benchmark images.*

## 5. Discussion, Limitations and Research Scope

While the coarse-to-fine initialization consistently improves paired PSNR across all 20 trials, a rigorous scientific evaluation requires emphasizing several key boundaries and trade-offs:

**1. Computational Trade-off:** The mean runtime for Coarse PnP is 0.3772 seconds, representing a 38.7% wall-clock overhead compared to 0.2719 seconds for direct DPIR. For time-critical inference, whether a +0.182 dB gain justifies a nearly 40% compute penalty depends heavily on the deployment context.

**2. Image-Dependent Sensitivity:** Performance gains are non-uniform across textures. Images with rich low-frequency structure and broad gradients (Coffee +0.397 dB, Astronaut +0.287 dB) benefit substantially. Conversely, high-frequency subjects (Chelsea +0.015 dB, Rocket +0.029 dB) exhibit marginal gains, suggesting that coarse representations discard fine edge information required for those textures.

**3. Restriction and Operator Approximation:** The 64 × 64 coarse inverse problem used an ad-hoc downsampled observation and scaled blur kernel rather than an exact mathematical restriction operator R satisfying $A_c = RA$. Developing consistent multilevel restriction and prolongation operators is essential for genuine multigrid convergence proofs.

**4. Scope Clarification:** This project constitutes an experimental exploration of initialization heuristics within DeepInverse. It does *not* implement a generative flow-matching model, nor does it claim theoretical convergence guarantees. It serves as a verified empirical foundation for future work on multilevel diffusion and flow-matching inversion [4].

## 6. Provenance and Reproducibility Protocol

All experimental artifacts are maintained in the repository under strict version control. Numerical data are stored in CSV format within results/, visual artifacts in figures/, and end-to-end execution routines in notebooks/DeepInverse_PnP_Image_Restoration.ipynb. Cryptographic checksums for all files are recorded in SHA256SUMS.txt to ensure experimental integrity and complete auditability.

## 7. Verified References

In accordance with rigorous academic citation standards, every reference listed below was independently verified against primary publisher databases and DOI registries during this session.

[1] J. Tachella, M. Terris, S. Hurault, A. Wang, D. Chen, M. Nguyen, M. Song, T. Davies, L. Davy, J. Dong, P. Escande, J. Hertrich, Z. Hu, T. I. Liaudat, N. Laurent, B. Levac, M. Massias, T. Moreau, T. Modrzyk, B. Monroy, S. Neumayer, J. Scanvic, F. Sarron, V. Sechaud, G. Schramm, C. Tang, R. Vo, and P. Weiss, "DeepInverse: A Python package for solving imaging inverse problems with deep learning," *arXiv preprint arXiv:2505.20160*, 2025.

*% verified: https://arxiv.org/abs/2505.20160, checked 2026-10-30*

[2] K. Zhang, Y. Li, W. Zuo, L. Zhang, L. Van Gool, and R. Timofte, "Plug-and-Play Image Restoration with Deep Denoiser Prior," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 6360-6376, Oct. 2022, doi: 10.1109/TPAMI.2021.3088914.

*% verified: https://ieeexplore.ieee.org/document/9446944, checked 2026-10-30*

[3] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, "Plug-and-Play Priors for Model Based Reconstruction," in *2013 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, pp. 945-948, Dec. 2013, doi: 10.1109/GlobalSIP.2013.6737048.

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[4] S. Martin, A. Gagneux, P. Hagemann, and G. Steidl, "Plug-and-Play Image Restoration with Flow Matching," in *International Conference on Learning Representations (ICLR)*, 2025.

*% verified: https://openreview.net/forum?id=V2c2YxK0eG, checked 2026-10-30*